SIMATS ENGINEERING

ASSESSMENT TOOL 3: Technical Presentation

COURSE NAME: Cloud Computing and Big Data Analytics Course Code CSA 15

COURSE OUTCOME COVERED

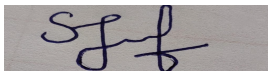
CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 10% **Total Marks:25**

Dave's Taxonomy: Imitation (Level 1)

Code of Conduct

I S Jenefar !92511345certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

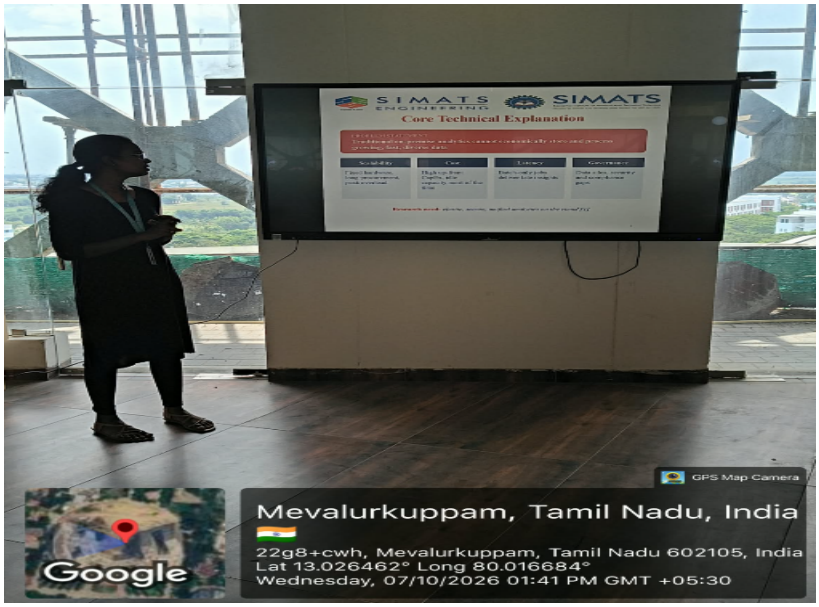
Assessment Tasks

Big Data Analytics in Cloud Computing

Explore cloud-based data processing architectures, analytics platforms, business value, and real-world applications.

Minimum slide flow:

- Title and objective
- Core technical explanation
- Architecture or process diagram
- Real-world use case
- Conclusion and key takeaway



BIG DATA ANALYTICS IN CLOUD COMPUTING

1. ABSTRACT

Big Data refers to datasets characterized by huge **volume, velocity, and variety**, which are difficult to store and process using traditional systems. Conventional on-premise analytics systems are costly, difficult to scale, and require significant hardware maintenance.

Cloud computing provides elastic storage, scalable computing resources, and managed analytics services that enable organizations to process large amounts of data efficiently. A cloud-based big data architecture generally includes data sources, data ingestion, data lakes, batch and stream processing, data warehouses, business intelligence, and machine learning.

Major cloud platforms such as **Amazon Web Services (AWS), Microsoft Azure, and Google Cloud** provide services for storage, processing, streaming, analytics, and machine learning. Cloud-based analytics can reduce infrastructure costs, provide faster insights, support large-scale data processing, and improve decision-making.

A real-world example is **Netflix**, which uses a Kafka-based event pipeline on AWS to process large volumes of user and application events for recommendations, playback quality, and operational analytics.

2. AIM AND OBJECTIVES

Aim

To understand how cloud computing enables scalable and cost-effective big data analytics.

Objectives

1.To understand cloud-based big data architectures and data-processing layers.

2.To study data processing pipelines used in cloud environments.

3.To compare analytics services provided by AWS, Microsoft Azure, and Google Cloud.

4.To analyze the business value of cloud-based big data analytics.

5.To study a real-world application of big data analytics using the Netflix case study.

3. INTRODUCTION

Big Data refers to datasets that are too large, fast, or diverse to be efficiently handled using traditional data-processing tools. Big data can be generated from various sources such as **IoT devices, social media, application logs, web activity, and business transactions**.

Cloud computing provides on-demand computing and storage resources through the internet. It follows an elastic and pay-as-you-go model, allowing organizations to increase or decrease resources according to their requirements.

The combination of cloud computing and big data analytics allows organizations to store and analyze large datasets without owning and maintaining expensive hardware infrastructure.

3.1 Five Vs of Big Data

Volume

Represents the enormous amount of data generated and stored, ranging from terabytes to petabytes.

Velocity

Represents the speed at which data is generated, transmitted, and processed, including real-time data streams.

Variety

Represents different forms of data, including structured, semi-structured, and unstructured data.

Veracity

Represents the quality, accuracy, and trustworthiness of data.

Value

Represents the useful insights obtained from data that can support better business decisions.

4. PROBLEM STATEMENT

Traditional on-premise analytics systems face several challenges when processing growing and diverse datasets.

Scalability

Traditional systems use fixed hardware capacity. Increasing capacity requires additional hardware and long procurement processes. During peak workloads, the existing infrastructure may become overloaded.

Cost

On-premise systems require high initial capital expenditure for hardware, storage, and infrastructure. Resources may remain unused during low-demand periods.

Latency

Traditional batch-based processing can produce insights only after hours or days, making it difficult to respond to real-time events.

Governance

Data may be distributed across different systems, creating data silos and increasing security, compliance, and management challenges.

Therefore, there is a need for an **elastic, secure, and unified cloud-based analytics environment**.

5. EXISTING SYSTEM

The traditional big data analytics environment commonly uses **on-premise Hadoop clusters and enterprise data warehouses**.

The infrastructure is purchased and maintained by the organization. Since the capacity is fixed, the system may struggle during peak workloads and remain underutilized during normal periods.

A typical traditional pipeline consists of:

Data Sources → Batch ETL → On-Premise Data Warehouse → Static Reports

The major limitations are:

High initial infrastructure cost.

Fixed processing and storage capacity.

Difficult scaling during peak workloads.

Long batch processing times.

Delayed or stale insights.

Data silos.

6. PROPOSED SYSTEM

The proposed system is a **cloud-native big data analytics platform** based on managed, elastic, and pay-per-use cloud services.

6.1 Elastic Storage

A cloud data lake can store extremely large datasets using object storage. Storage capacity can scale to petabyte levels according to requirements.

6.2 Managed Processing

Cloud platforms provide managed batch and stream-processing engines. Organizations can process data without maintaining the underlying processing clusters.

6.3 Serverless Data Warehouse

Serverless warehouses provide SQL-based analytics without requiring users to manage physical infrastructure. Users can pay according to their usage.

6.4 Self-Service BI and Machine Learning

Business intelligence dashboards can provide visual insights, while machine learning and AI services can be used for prediction and advanced analytics.

6.5 Security and Governance

Cloud platforms support encryption, access control, auditing, monitoring, and governance mechanisms to protect organizational data.

6.6 Pay-As-You-Go Model

Cloud resources can be increased during high-demand periods and reduced during low-demand periods. This helps reduce unnecessary infrastructure costs.

7. OVERALL SYSTEM ARCHITECTURE

The proposed architecture consists of multiple loosely coupled layers.

7.1 Data Sources

Data can originate from:

- IoT devices and sensors
- Application
- System logs
- Clickstream data
- Social media
- Websites
- Databases

7.2 Data Ingestion

The collected data is transferred into the cloud using

- Batch uploads
- Streaming systems
- APIs
- Change Data Capture (CDC)

Examples include **Kafka and Amazon Kinesis**.

7.3 Storage

The data is stored in cloud-based data lakes and lakehouses.

Examples:

- Amazon S3
- Azure Data Lake Storage Gen2
- Google Cloud Storage
- Delta Lake
- Apache Iceberg

Data can be organized into raw and curated zones.

8. PROPOSED METHODOLOGY

The cloud analytics pipeline converts raw data into useful information through the following stages:

Step 1: Ingest

Data is collected from different sources and transferred to the cloud.

Step 2: Store

The collected data is stored in a cloud data lake or other cloud storage system.

Step 3: Clean and Transform

Raw data is cleaned, organized, transformed, and prepared for analysis.

Step 4: Analyze

Analytics algorithms and processing engines are applied to extract useful information.

Step 5: Visualize and Act

The results are presented through dashboards, reports, predictions, or other decision-support systems.

9. MAPREDUCE ALGORITHM

The proposed methodology uses the **MapReduce approach on Apache Spark** for parallel data processing.

Map

The input data is divided into smaller partitions or blocks. Different workers process these partitions simultaneously.

Shuffle and Sort

The intermediate records generated by the Map stage are grouped according to their keys.

Reduce

The grouped records are aggregated to generate the final results.

Output

The processed results are stored in cloud storage or passed to further analytics systems.

Advantages

Parallel processing of large datasets.

Fault tolerance

Automatic re-execution of failed tasks

Elastic scalability.

Efficient processing of large volumes of data

10. CLOUD ANALYTICS PLATFORMS

Layer	AWS	Microsoft Azure	Google Cloud
Storage	Amazon S3	ADLS Gen2	Cloud Storage
Batch / ETL	EMR, Glue	Databricks, Data Factory	Dataproc, Dataflow
Streaming	Kinesis, MSK	Event Hubs	Pub/Sub, Dataflow
Warehouse	Redshift	Synapse Analytics	BigQuery
BI & ML	QuickSight, SageMaker	Power BI, Azure ML	Looker, Vertex AI

The choice of cloud platform depends on the organization's existing technology stack, available skills, pricing requirements, and compliance needs.

11. BUSINESS VALUE

11.1 Lower Cost

Cloud computing reduces the need for large upfront infrastructure investments. Organizations can move from a capital expenditure model to an operational expenditure model and pay according to usage.

11.2 Faster Insights

Real-time data streams and self-service analytics help organizations reduce the time between data generation and decision-making.

11.3 Scale and Agility

Cloud platforms can support very large datasets and allow organizations to scale their resources according to workload requirements.

11.4 Smarter Decisions

Big data analytics supports applications such as personalization, fraud detection, forecasting, machine learning, and AI-based decision-making.

Overall, business value can be achieved through **faster decisions, lower infrastructure costs, and data-driven products and services**.

12. REAL-WORLD USE CASE: NETFLIX ON AWS

Netflix is a real-world example of large-scale cloud-based big data analytics.

Netflix uses a **Kafka-based event pipeline on AWS** to process events generated by applications and devices.

The system handles extremely large volumes of data. The presentation describes approximately **500 billion events per day in 2016**, around **1.3 petabytes of data per day**, and approximately **8 million events per second at peak**.

Data Flow

Applications & Devices → Kafka → Routing Service → S3 / Elasticsearch → Analytics

Applications and Devices

Playback events, user interface events, and other application activities generate data.

Kafka

Kafka buffers and handles the incoming event streams.

Routing Service

The routing service distributes the collected data to appropriate storage and processing systems.

S3 / Elasticsearch

Data is stored and searched using cloud-based storage and search technologies.

Analytics

Tools such as Hive, Spark, and EMR process the data and generate useful insights.

These insights support:

Content recommendations

Playback quality improvement

Operational monitoring

Data-driven decision-making

The Netflix example demonstrates how cloud-based big data analytics can operate at very large scale

15. CONCLUSION

Traditional big data analytics systems are often costly, rigid, and difficult to scale as data volume, velocity, and variety increase. Cloud computing provides an effective solution through elastic storage, managed processing, serverless analytics, and pay-as-you-go resources.

A cloud-based architecture can combine data lakes, batch and stream processing, data warehouses, business intelligence, and machine learning into a unified analytics environment.

AWS, Microsoft Azure, and Google Cloud provide a wide range of services for implementing such systems. The Netflix case study demonstrates the practical use of cloud-based big data analytics at large scale.

Therefore, cloud computing makes big data analytics **more scalable, cost-effective, flexible, and accessible**, enabling organizations to obtain faster insights and make better data-driven decisions.

16. REFERENCES

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